Granular Income Inequality and Mobility using IDDA: Exploring Patterns across Race and Ethnicity

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Abstract

Shifting earnings inequality among U.S. workers over the last five decades has been widely studied, but understanding how these shifts evolve across smaller groups has been difficult. Publicly available data sources typically only ensure representative data at high levels of aggregation, so they obscure many details of earnings distributions for smaller populations. We define and construct a set of granular statistics describing income distributions, income mobility and conditional income growth for a large number of subnational groups in the U.S. for a two-decade period (1998-2019). In this paper, we use the resulting data to explore the evolution of income inequality and mobility for detailed groups defined by race and ethnicity. We find that patterns identified from the universe of tax filers and W-2 recipients that we observe differ in important ways from those that one might identify in public sources. The full set of statistics that we construct is available publicly as the Income Distributions and Dynamics in America, or IDDA, data set.

Keyword*:* income inequality, racial income disparities, gender wage gap, granular inequality

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[*] The opinions and conclusions expressed here are those of the authors should not be interpreted as reflecting the views of the U.S Census Bureau, the Federal Reserve Board of Governors, the Federal Reserve Bank of Minneapolis, or any other person associated with the Federal Reserve System. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7511151; Disclosure Authorization Numbers CBDRB-FY23-0277, CBDRB-FY23-0373, CBDRB-FY23-CES014-019, and CBDRB-FY23-CES014-016.)

1 Introduction

Earning power in the United States is significantly more unequal today than it was fifty years ago, when a multi-decade pattern of widely shared growth began to retreat. Since 1980, the pace of earnings growth has been uneven, and varying, for Americans throughout the earnings distribution. The course of the widening earnings disparities has been well-documented, and a number of patterns and episodes emerge (Hoffmann et al., 2020). The highest earners have experienced consistent growth in their earning power since 1980, although this has slowed or become more concentrated among the very highest earners since about 2000 (Kopczuk et al., 2010; Blanchet et al., 2022). Relative real earnings for workers below the median have fallen and partially recovered more than once (Autor et al., 2008; Blanchet et al., 2022; Autor et al., 2023). Although these patterns are robust, important differences across demographic groups, notably between women and men, have been documented (Guvenen et al., 2022).

In spite of this significant body of evidence, there is still much we do not know about how earnings have changed for many Americans over the last five decades of stubbornly rising inequality. The open questions are particularly acute at the subnational level. For example, how has income inequality evolved in each of the broad range of race and ethnic groups in America? Have U.S. groups defined by race and ethnicity experienced the major episodes in changing income inequality in similar ways, or not? Do patterns in income inequality for U.S. demographic groups systematically vary across U.S. states? There are also many questions about how income evolves over time for members of these groups that have been difficult to answer. Do low earners in all demographic groups move into higher earnings percentiles at the same rate? And again, do these patterns differ depending on the state where individuals live and work?

Answers to such questions have been limited in part by data availability. Even large surveys like the American Community Survey (ACS) capture samples that are small in absolute terms for groups that constitute small shares of the population (either nationally or in a specific place), and these small samples may not support reliable estimates that describe the full distribution of income. Surveys also tend not to follow respondents over time, making it difficult to use them to understand income dynamics, or else trade off cross-sectional sample size against panel duration, limiting the granularity of any income dynamics estimates that may be producible. Moreover, measuring top

incomes well is critical for understanding inequality, but measures based on surveys can be distorted by both disclosure avoidance procedures (e.g. topcoding) and survey misreporting (Burkhauser et al., 2011; Bee and Mitchell, 2017).

Such data issues are particularly important when it comes to understanding differences in income inequality and dynamics across groups defined by race and ethnicity, as these groups are unevenly distributed across space and differentially likely to be affected by measurement issues related to top incomes. It is especially important to be aware of differences in the prevalence of groups across space in this context, as state income distributions have become less similar to each other at the top in recent decades, and this phenomenon appears to be driven by the top of the White income distribution (Rinz and Voorheis, 2023).

Answering questions like these requires data on income distributions and income mobility at demographically granular levels. We define a data source as having this granularity if measures of income distributions and dynamics are available in three dimensions: for many demographic groups, for geographic areas below the national level, and for intersections of the previous two groups. Observing the evolution of income distributions at these levels is key for more fully understanding and, potentially, addressing US earnings inequality.

To facilitate this kind of analysis, we developed and published the Income Distribution and Dynamics in America data set, or IDDA. IDDA is built from the universe of tax records matched to internal Census Bureau demographic information for the period 1998 to 2019. It reflects the experiences of the populations of U.S. tax filers and W-2 employees over this period. Broadly, we tabulate statistics from individual-level tax records to describe the distribution of incomes and movements through it over time for a large set of demographic groups. More precisely, we compute statistics for groups defined by four elements: income source and unit; U.S. or individual states; year; and demographic groups defined by age bins, sex, race, ethnicity, birthplace (U.S. or abroad), and intersections of the demographic groups. Within each group, we calculate two types of statistics. The first type characterizes the distribution of incomes for a group in several ways. These include percentile values ranging from the 10th percentile to the top 99.999th percentile for the group; shares of income going to top earners; and shares of the group among all top earners at a point in time. The second type characterizes the mobility of incomes in the group across income distribution bins over time as well as the distribution of income changes experienced starting from

a given initial income distribution bin.

Although IDDA has many potential applications, in this paper, we use it to answer two specific questions. First, how do patterns in the distribution of earnings over the two decades leading up to the global Covid-19 pandemic in IDDA compare to patterns estimated using publicly available sources based on survey samples? We begin by comparing patterns in income and earnings inequality for U.S. household and workers overall before turning to the demographic subgroups that motivated our granular approach to constructing IDDA. Specifically, we document patterns of income inequality for demographic sub-groups defined by detailed race and ethnicity identities, along with state-level location using IDDA and ask how these compare to patterns for the same groups constructed from survey samples. Since the main public-use surveys of earnings do not guarantee precision at the level of our IDDA groups, there is reason to suspect patterns will differ. We use these comparisons to discuss when, where, and how various groups depart from the main patterns in overall U.S. inequality.

Our second question asks, what patterns in earnings dynamics accompanied the development of inequality that we observed? This analysis leverages the second major set of IDDA statistics, those describing income mobility and changes for groups. Specifically, how do changes in the values of various percentiles across the income distribution compare with changes in the incomes of specific people in similar parts of the distribution? Do those within-person changes in income differ across groups defined by race and ethnicity?

This paper and the data construction behind it contribute to a growing field in which administrative sources are leveraged to understand detailed distributional patterns in the U.S. economy while also sharing aggregate data so that other researchers may more quickly innovate. Such efforts often include research papers and data repositories. Chetty et al. (2014) analyze measures of inter-generational income mobility by location of residence for a cohort of individuals born in the early 1980s and made their data available via their Opportunity Atlas portal Chetty et al. (2018). Blanchet et al. (2022) analyze shares of income and economic growth accruing in the most recent daily period to U.S. earners by percentile, and they make their method available on their Real-Time Inequality site. A forthcoming issue of Quantitative Economics, multiple papers explore detailed country-level inequality data constructed from administrative data from 13 countries and released

as the Global Research in Income Dynamics (GRID) Repository.[2]

Relative to these efforts, IDDA offers several advantages for studying subnational patterns in labor market earnings and incomes for its populations of U.S. tax filers and W-2 employees. The major advantages of our approach are threefold. First, our data include a wide range of statistics for subnational geographic areas (states) as well as many subnational and sub-state populations. Second, we combine measurement of income mobility and changes with measurement of income distributions from a single source for our populations of interest. And third, the statistics that underlie our analysis are computed directly from the universe of tax information, which means we do not need to rely on modelling assumptions to link micro and macro data sources.

These features are especially important for producing granular measures characterizing experiences across income distributions and income dynamics across groups defined by place, income rank, and demographic groups such as gender, age, U.S.-birth status, race and ethnicity, or their intersections. Measuring such experiences can be very demanding in terms of the number of observations required to produce them, and relying on samples generated by even large surveys like the ACS can leave some groups without estimates for most places and most groups without estimates for some places. Combining survey and administrative data as we do in this project substantially expands the set of groups and places that can be included, as well as the possible measures that can be produced. Akee et al. (2019) use these linkages at the US level to document differences across race/ethnicity groups in incomes, top income shares, and income mobility. Rinz and Voorheis (2023) also leverage these linkages at the subnational level for tax units to discuss large differences in income growth across race/ethnicity groups within a state, especially for top incomes.[3]

Our approach has many strengths, but the measures constructed for IDDA have some limitations. These stem from limitations in the populations and types of income observed. To answer our questions of interest, we examine the universe of U.S. tax filers and W-2 recipients. These are certainly important groups, but they not representative of all U.S. workers or households and we did not attempt to reweight the data to reflect other important populations of interest. Also, the income from which our statistics are tabulated is essentially limited to pre-tax and transfer sources. This means it is inadequate for studying questions in which income after redistribution is a key

[2]See https://mebdi.org/global-repository-income-dynamics.

[3]See Partridge et al. (1996); Morrill (2000); Frank (2014); Sommelier and Price (2014) for existing work on subnational inequality in the U.S.

component, for example, questions related to poverty.

This paper proceeds as follows. We describe IDDA's construction in Section 2. In section 3, we use IDDA to analyze trends in earnings inequality for detailed race and ethnicity groups, and we compare these to what might be inferred from the Current Population Survey (CPS). Section 4 describes patterns in income mobility and conditional income changes for the same detailed groups in IDDA. In Section 5, we ask similar questions to those in previous sections but we provide analysis for groups and places that are not readily available in any public sources: Native Americans on Census-defined Native Lands. Section 6 concludes.

2 IDDA Construction

The Income Distributions and Dynamics in America (IDDA) dataset consists of novel granular statistics on the distribution of incomes and income changes as well as income mobility within cells defined by demographics, income rank, place, and time. This dataset is generated from underlying microdata that combine income and geographic information from the universe of individual tax returns with demographic information available from various sources through the U.S. Census Bureau. The key features of these data are difficult to collectively replicate in other settings: they provide detailed demographic information, high-quality measurements of top incomes, and statistically meaningful sample sizes.

Income types and primary samples in IDDA: We report select income distribution percentiles, select income change distribution percentiles, and income mobility statistics within two primary tabulation samples: (i) a household-level income dataset aggregated from Form 1040s, and (ii) individual-level earnings dataset built from information reported on Form W-2s.

The W-2 sample covers tax years 2005 to 2019 and includes estimates of individual wage/salary compensation, deferred compensation, and total compensation (their sum). Wage/salary compensation is reported in Box 1 of Form W-2 and is positive for all earners included in the sample. Deferred compensation is recorded in Box 12a-12d and may be positive or zero. Total compensation is the primary W-2 earnings measure available in IDDA at both the national level and for individual states.

The IRS 1040 sample converts tax filing-unit income information reported on Form 1040 into measures of household resources available to individuals in particular demographic groups. To construct households, we leverage linkage to the Census Bureau Master Address File ID (MAFID). We circumvent issues associated with the notion of a 'household head' or the assumptions required for the apportionment of household income among its members. Instead, we aggregate income values across all 1040 forms filed at a shared address and assign these household totals to each member of the common address. While this concept does not neatly lend itself to distributional national account applications, it allows us to preserve the rich variety of demographic characteristics within a household. In doing so, we are able to document top incomes at the household level and to contrast their levels and fluctuations by race/ethnicity, its intersections with gender and age, and by state.[4]

We apply this aggregation method to measures of earned and unearned income obtained from Form 1040: adjusted gross income (AGI), wage and salary income, and non-wage income defined as AGI minus household wage and salary income. In addition to wage and salary income, AGI includes schedule C self-employment income, interest and dividends, and taxable retirement income, minus deductions (for a more detailed description of the components of AGI, see Appendix A). We produce estimates based on these measures for 1998 through 2019.

Demographic characteristics: We leverage the assignment of the Census Bureau's unique, anonymized protected identification key (PIK) to all data files we use (Wagner and Layne, 2014). As in Akee et al. (2019), this allows us to link information on race, ethnicity, gender, age, foreign born status, and other demographic characteristics to individual tax records. As discussed above, this is a particularly important feature especially when it comes to minority racial and ethnic groups and their intersections with gender or age.

Our preferred source for determining race and ethnicity is the Census Bureau's 2020 Best Race and Ethnicity Administrative Records Composite File, which uses a variety of sources including records from the Department of Housing and Urban Development, the Temporary Assistance for Needy Families program, the Indian Health Service, and other agencies to assign a single race and Hispanic ethnicity to individuals in the Census Bureau data system. If a record does not appear in

[4]See for example Guvenen and Kaplan (2017) for a discussion on units the issue of individual records, tax-filing-unit records, and household units when it comes to top income and inequality.

the Best Race and Ethnicity Administrative Records Composite File, we use the most recent race and ethnicity data from either the Decennial Census (2000 or 2010) or American Community Survey (2005-2019), if available. The large size of the IRS dataset allows us to disaggregate statistics for six major race and ethnicity groups: Hispanic, non-Hispanic Asian, non-Hispanic American Indian and Alaska Native (AIAN), non-Hispanic Black, non-Hispanic Native Hawaiian or Pacific Islander (NHPI), and non-Hispanic White. At the U.S. level, statistics are also reported for the non-Hispanic other or multiple-race group. Sex, place of birth, and year of birth and death (which determine age) are drawn from Social Security Administration records. State of residence is taken from Form 1040 or any available information returns.

Individuals are excluded from the cross-sectional data if we are unable to link key demographic or geographic information or if tax records indicate conflicting information. We also exclude individuals whose address ID is shared by an unusually large number of records. Otherwise, we impose minimal sample restrictions.

The PIK also facilitates longitudinal linkage, and we use the panel dimension of the tax data to document the distribution of income growth and mobility at different initial income levels. This further divides small demographic cells into income brackets. We also report statistics using a secondary W-2 sample restricted to prime-age workers with earnings above a threshold, equivalent to working half-time for 13 weeks at the federal minimum wage.

IDDA modules and availability of statistics: We produce a variety of statistics characterizing income distributions within years as well as conditional income changes and mobility across pairs of years. Table 1 lists these categories of statistics, which we term modules, as well as the demographic characteristics we use to define groups. We build on Akee et al. (2019) by constructing yearly state-level statistics on income inequality by race, by race and age, and by race and gender for different types of income. These measures include group-percentile income values, top income shares within demographic groups, and the share of top income held by members of different demographic groups. We also build on the contribution of McKinney et al. (2022) in the Global Repository of Income Dynamics (GRID) project by documenting percentiles of 1-year income changes and 5-year income changes by initial income level, type of income, by demographic group, and across states. For a more extensive definition of the statistics included in each module of IDDA, refer to the technical

Table 1: Summary of statistics in IDDA

Statistics		Demographic characteristics defining groups
Income levels	Transition matrices and income change distributions	
Percentile values: p10, p25, p50, p75, p90, p95, p98, p99, p99.9, p99.99, etc. as sample size allows	**Transition matrices**: $P(B_{y+k}\|B_y)$ for k={1,5} and income bins B={missing, p0-p24, p25-p49, p50-p74, p75-p89, p90-p100}	**Race/ethnicity**: Hispanic, Non-Hispanic Black, Non-Hispanic AIAN, Non-Hispanic Asian, Non-Hispanic NHPI, Non-Hispanic White
Top income shares (within group): of all income earned by a given group, what share was earned by the top 10%, 5%, 2%, 1%, .1%, .01%, etc. of earners, as sample size allows	**Conditional income changes**: mean, p10, p25, p50, p75, and p90 of nominal income changes for individuals starting in each income bin B between y and y+k	**Age**: 16-24, 25-34, 35-44, 45-54, 55-64, 65+
Top income shares (across groups): of income earned by the top 10%, 5%, 2%, 1%, .1%, .01%, etc. of all earners (as sample size allows), the share earned by members of a given group		**Gender**: male, female
Top income population shares: The share of the top 10%, 5%, 2%, 1%, .1%, .01%, etc. of earners (as sample size allows) belonging to a given group		**Place of birth**: U.S.-born, foreign born

Note: p# indicates percentile value. Top income percentiles used for state-level inequality statistics go up to the 98th percentile. AIAN = American Indian and Alaska Native. NHOPI = Native Hawaiian or Other Pacific Islander. Top income bins in transition matrices and conditional income changes are consolidated into a single p76-p100 bin for estimates at the state level. Estimates are produced nationally and at the state level within the groups listed for each demographic characteristic, as well as for two-way interactions of select demographic characteristics.

documentation in Appendix A.

All statistics in IDDA are calculated within a sample and income concept, year, and geography, and may be defined within or across demographic groups. The demographic groups included in the dataset vary across the different statistics modules and by sample and geography, in part to manage the population size underlying each statistic. While IDDA is constructed from large data, the granular information it provides can still lead to small samples and some statistics are suppressed as a result.

Table 2 summarizes the availability of statistics in each of these three categories– income levels, transition matrices, and conditional income changes– by sample and geography. Availability is simply the percent of all defined statistics that are available (not suppressed) given the demographic disaggregations reported in a particular sample and geography. In general, availability is low at the

intersections of age and race, which is reported in the U.S. and state W-2 income levels, U.S. W-2 transition matrices and income change distributions, and U.S. 1040 income levels. The transition matrices have low coverage in the prime-age working sample because individuals can "age in" or "age out" of the sample over time, leading to a number of statistics that are defined for small populations.

Table 2: Available IDDA statistics by geographic level and by income sample

	U.S.-level		State-level	
	1998–2004	2005–2019	1998–2004	2005–2019
Household-1040				
Income Levels	86%	88%	97%	98%
Transition Matrix	100%	100%	97%	97%
Income Changes	100%	100%	97%	98%
Individual-W2				
Income Levels	–	87%	–	85%
Transition Matrix	–	97%	–	95%
Income Changes	–	$> 99\%$	–	99%

Note: Columns provide availability rates across all demographic groups included in each module, geography, and sample. The W-2 sample covers the 2005-2019 period only. For a full list of the demographic disaggregations included in each module, refer to the technical documentation in Appendix A.

Table 3 provides a more detailed analysis of availability for individual race/ethnicity subgroups. Availability is given as a percent of all defined statistics pertaining to a specific subgroup, including intersections of age and race and race and sex. For example, statistics defined for the groups Asian Male, Asian Female, and Asian 25-34 each count as 1 in the denominator of the availability rate for the non-Hispanic Asian group.

Coverage falls for small race/ethnicity groups as more intersections are included and statistics are reported further into the tail of the income distribution. In particular, statistics are often suppressed for the Native Hawaiian/Pacific Islander and American Indian or Alaska Native groups within individual U.S. states. Still, availability is quite high at the national level across all groups. The fact that some suppression is still required emphasizes the importance of preserving population

coverage of the tax data.

Table 3: Available IDDA statistics by race and income samples

	Intersections	Hispanic	AIAN	Asian	Black	NHOPI	White
U.S. Household-1040							
Income Levels	age	92%	76%	89%	88%	69%	95%
Transition Matrix	–	100%	100%	100%	100%	100%	100%
Income Changes	–	100%	100%	100%	100%	100%	100%
U.S. Individual-W2							
Income Levels	age, sex	90%	74%	88%	91%	66%	92%
Transition Matrix	age, sex	100%	95%	100%	100%	83%	100%
Income Changes	age, sex	100%	100%	100%	100%	98%	100%
U.S. Prime-age working-W2							
Income Levels	age, sex	94%	77%	91%	93%	71%	94%
Transition Matrix	age, sex	79%	76%	79%	78%	69%	79%
Income Changes	age, sex	100%	100%	100%	100%	100%	100%
State -1040							
Income Levels	–	100%	97%	100%	98%	72%	100%
Transition Matrix	–	100%	95%	99%	98%	62%	100%
Income Changes	–	100%	97%	100%	99%	74%	100%
State-W2							
Income Levels	age, sex	91%	76%	89%	88%	44%	95%
Transition Matrix	–	97%	84%	94%	94%	47%	100%
Income Changes	–	100%	98%	100%	100%	78%	100%

Note: Cells report availability rates as defined in Table 2. The column "intersections" indicates whether statistics in the given geography/sample are provided for the intersection of age and race, race and sex, or neither. In general, statistics are provided further into the tail of the income distribution of the U.S.-level data than the state-level data. Income levels are reported up to the 98th percentile at the state level and 99.999th percentile nationally. Transition matrices and conditional income changes are calculated within quartiles of the state-level income distributions. At the U.S. level, the top income quartile is divided into two smaller brackets (the 75th-90th percentile and above the 90th percentile). Rates are averages across all years.NHOPI abbreviates Native Hawaiian or Pacific Islander; AIAN abbreviates American Indian or Alaska Native

3 Evolution of Income Inequality for Detailed Demographic Groups, 1998-2019

We begin by plotting changes in earnings, by percentile, over time for the population of U.S. tax filers as a whole. Panel (a) in Figure 1 graphs the log difference (or approximate percentage change) in percentile values for household earnings relative to their values in 1998 using IDDA. Broadly, this produces a familiar pattern. Since, 1998, household earnings have grown but earnings for top percentiles have grown more. By 2018, earnings for a household at the 90th percentile of household earnings had risen 30 percent relative to their 1998 level. Earnings for a household at the 25th percentile increased by less than half that, and earnings for the lowest earning households (10th percentile) had not grown at all. Other episodes familiar from earlier studies of U.S. inequality are also apparent in the IDDA data. The period of comparatively broad-based growth in earnings in the latter half of the 1990s is apparent at the beginning of the IDDA series. Earnings declines and their slow recovery following the Great Recession - particularly for the lowest earning households - is also apparent in the IDDA series.

The IDDA series suggest that changes in earnings percentiles unfolded relatively smoothly year to year with some notable exceptions. First, the IDDA series show that earnings at the lowest percentiles can change sharply and temporarily, as they did in 2007 and 2019. These are likely driven by filing changes.[5] Sharp changes in the value of earnings at the very highest levels - the 98th percentile, for example - also occur in the IDDA series, but these likely reflect large fluctuations in compensation for the very highest earners that accompany recessions.

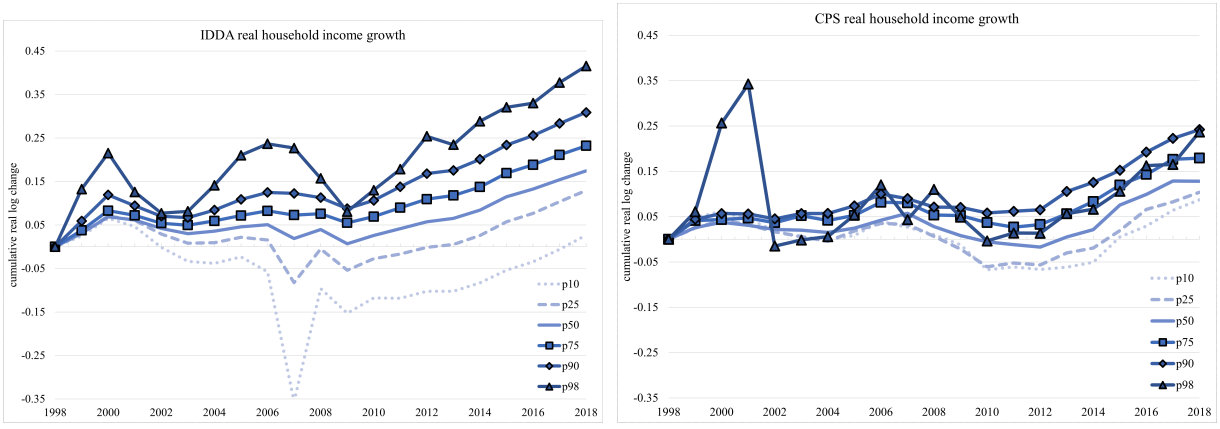
In Panel (b), we produce the same figure using publicly available data from the Annual Social and Economic supplement (ASEC) to the CPS. We choose income concepts reported in IPUMS CPS that align well with the concepts measured in our administrative data Flood et al. (2022).[6] We compute analgous statistics in public CPS microdata to those we analyze in IDDA, imposing only the restriction that CPS cells must contain at least 50 observations in order for us to include a CPS-derived statistic in our analysis.

At the most general level, the two sources tell a similar story: earnings inequality has widened

[5] Extensions of the paper will discuss how to interpret changes in light of potential filing changes.
[6] Specifically, these are HTOTALVAL for household AGI and WSAL˙VAL for W-2 earnings.

since the late 1990s, with recessions slowing income growth more sharply (or even eroding income) for the lowest earners. Both series also show that earnings growth in recoveries has been slow to return lower earners to pre-recession levels, and the late 1990s remain the last period in which the U.S. experienced broad-based income growth that was matched across earnings levels. Despite similarities, the CPS figure shows that - even at this highly aggregated level - the CPS produces patterns that suggest it may be an unreliable measure for some groups. In the highest earning bin shown (the 98th pecentile), the CPS series fluctuates widely, and it does not reproduce a pattern of higher earnings growth for the highest percentiles that has been documented elsewhere.

Figure 2b repeats the same exercise using the W-2 earnings construct available from 2005 onwards. At a high level, the two sources again tell a broadly similar story to one another. Both also show that the final years of recovery from the Great Recession saw greater growth in wage earnings for the lowest earners, even though this did not translate to greater income growth for the lowest earning households. But again, even at this aggregate level, the series show meaningful differences. First, for the lowest wage earners (at the 10th and 25th percentiles), IDDA appears to more accurately identify the beginning of earnings recovery for them. And second, for the highest wage earners, IDDA shows a pattern that seems more in line with changes for other above-median wage earners than does the CPS, which suggests that the highest earners saw some of the greatest earnings declines in the Great Recession and recovered no faster than most other groups.



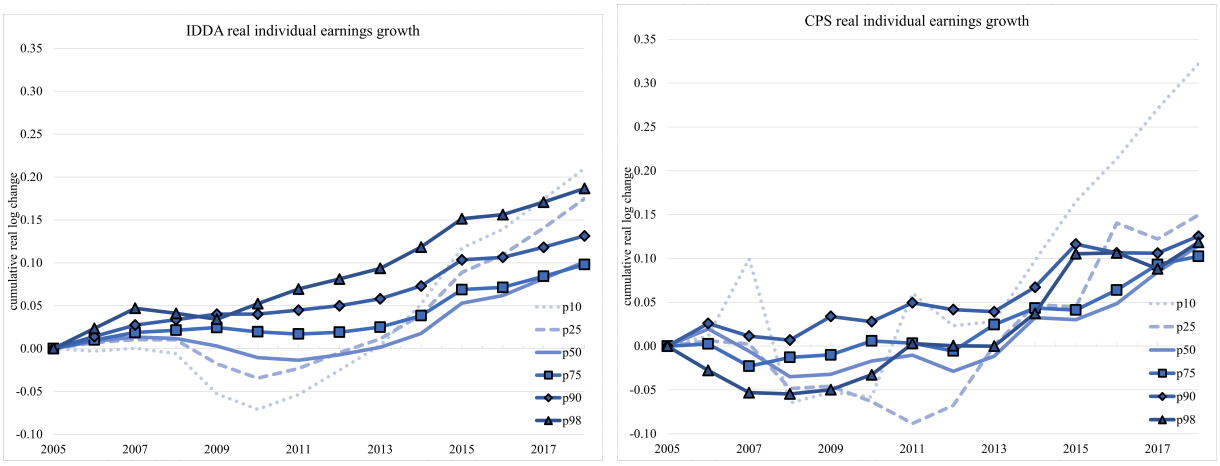
(a) IDDA household income growth

(b) CPS household income growth

Figure 1: Household income growth in IDDA vs CPS

Source: IDDA and authors' calculation using IPUMS CPS. Release authorization CBDRB-FY23-0277.

In Table 4, we quantify some of the changes that Figures 1 and 2 present visually. Specifically,



(a) Earnings growth

(b) Earnings income growth

Figure 2: Earnings growth in IDDA vs CPS

Source: IDDA and authors' calculation using IPUMS CPS. Release authorization CBDRB-FY23-0277.

we report the change in inequality measured by the 90-10 and 75-25 ratios in the two series for several data subperiods. The far right columns of the top panel confirm the impression from Figure 1: the IDDA series shows a greater increase in inequality between the very highest and lowest earning households over the period than does the CPS. This is true in both subperiods of the data. The middle two columns show that initial values for inequality - measured as the 90-10 ratio of household level earnings - were somewhat lower in IDDA (8.37) and higher in the CPS (8.93). Nevertheless, we consider the similarity in these measures across the two series to be notable given the major differences in underlying sample and method. The second panel of Table 2 shows that the two series produce even closer measures of the 75-25 ratio; the long differences in the 75-25 ratios are also similar across the series. The bottom panel shows that the two sources do not match as well when comparing the 90-10 and even (though to a lesser degree) the 75-25 ratios of wage and salary earnings at the start of the period for this series (2005).

It is worth noting that Figures 1 and 2, and Table 4, also suggest some material differences in the evolution of inequality in the IDDA series as compared to the CPS. Both IDDA series show a more monotonic expansion of inequality across earnings percentiles in the wake of the Great Recession than is observed in the CPS. The IDDA series also show a larger total expansion in inequality by the end of the period than does the CPS; this is true for household inequality overall and for wage earnings inequality above the median. These differences suggest potentially new insights into the

Table 4: Percentile ratios and changes in IDDA vs CPS

Period	Income	Ratio	Initial value IDDA	Initial value CPS	Change IDDA	Change CPS
1998–2004	HH Gross Income	p90p10	8.37	8.93	1.09	0.53
2005–2011	HH Gross Income	p90p10	9.55	9.53	1.26	0.56
2012–2018	HH Gross Income	p90p10	10.97	10.19	0.12	0.24
1998–2004	HH Gross Income	p75p25	2.88	3.02	0.14	0.15
2005–2011	HH Gross Income	p75p25	3.03	3.13	0.14	0.18
2012–2018	HH Gross Income	p75p25	3.22	3.31	–0.05	–0.03
2005–2011	Earnings	p90p10	27.95	15.00	2.89	–0.16
2012–2018	Earnings	p90p10	30.13	15.28	–4.29	–2.95
2005–2011	Earnings	p75p25	4.69	3.35	0.32	0.19
2012–2018	Earnings	p75p25	4.80	3.56	–0.37	–0.46

Source: IDDA and authors' calculation using IPUMS CPS. Release authorization CBDRB-FY23-0277.

evolution of U.S. inequality since the late 1990s. However, our aim in this paper is to assess IDDA as a source for patterns in earnings inequality for detailed sub-national groups, so we leave these questions for future work.

To begin to understand patterns in earnings inequality for populations defined by race, ethnicity and place, we extend the analysis in Table 4 to groups defined by these dimensions in IDDA and in the CPS. As in Table 4, we report results using the household level adjusted gross income in the top panels. Results using the full sample of W-2 wage and salary earners appear in the bottom panels. We divide each population into subgroups defined by race and ethnicity (as shown) and state of residence. We then compute the 90-10 ratios for each income concept at the beginning and end of its period of availability for each subgroup and regress the long-difference in this ratio in IDDA on the analogous long-difference in the CPS. We report the slope coefficients from this regression (estimated without a constant) along with its standard error and the regression's R-squared in the Table 5.

The rightmost two columns show the number of states for which we were able to compute a long-

difference in each source.[7] IDDA shows much better coverage for this basic measure of inequality than the CPS. Putting aside the question of the relative quality of estimates from each source, it is clear from Table 3 that it is simply not possible to study basic questions of inequality for many race and ethnic groups using CPS data at the state level.[8]

Turning to results from the regression estimates, there are several points to note. First, for large groups, IDDA and the CPS produce estimates of changes in the 90-10 ratios at the state level that are well-correlated with one another. The coefficient on the simple regression is 0.51 using household gross income and 0.65 using wage and salary earnings from full state-level populations; both are highly statistically significant. The correspondence between IDDA and CPS 90-10 measures is also strong at the state-level for non-Hispanic White populations. As we examine state-level populations for other race and ethnicity groups, this correspondence deteriorates considerably. For most subgroups, the relationship between 90-10 earnings inequality (under either income definition) as measured in IDDA is not significantly predictive of the CPS measure. This is likely due to insufficient data in the CPS to reliably estimate even these simple inequality measures at the state level. Low N values for the CPS is one manifestation of this problem. For the NHOPI groups, it is not even possible to create the 90-10 measure in the CPS for a single state.

Using a measure of dispersion from closer to the center of the distribution does not change the substantive conclusions from Table 5. In Table 6 we repeat this analysis using the 75-25 earnings ratios. Here IDDA is a somewhat better predictor of the CPS inequality measures for large groups. For example, the coefficient rises to 0.69 in the full samples using gross income. But standard errors on the slope coefficients show that the CPS measure and IDDA are not significantly related for almost any race or ethnicity group at the state level.

4 Income Dynamics for Detailed Demographic Groups: 1998–2019

Tracking changes in cross-sectional estimates of different percentiles of the income distribution can provide a sense of how growth varies across the distribution. However, because people do not

[7] For CPS calculations, we required that each state-group cell contain at least 50 observations.

[8] The ACS has sufficient sample size to provide precise cross-sectional measurement of inequality, but lacks the longitudinal structure of the CPS-ASEC or administrative records.

Table 5: 90/10 percentile ratio changes in CPS vs IDDA across states

Income	Group	Est. slope β	Std. Error	R. Sq.	N_{CPS}	N_{IDDA}
		$\Delta\left(\frac{\text{p90}}{\text{p10}}\right)^{\text{state}}_{\text{CPS}} = \beta\Delta\left(\frac{\text{p90}}{\text{p10}}\right)^{\text{state}}_{\text{IDDA}} + \varepsilon^{\text{state}}$				
HH Gross Income	All	0.51	0.07	0.52	51	51
HH Gross Income	White	0.46	0.09	0.36	51	51
HH Gross Income	Hispanic	0.02	0.43	0.00	32	51
HH Gross Income	Black	2.48	0.98	0.18	30	51
HH Gross Income	Asian	–	–	–	0	51
HH Gross Income	AIAN	1.73	3.97	0.03	7	51
HH Gross Income	NHOPI	–	–	–	0	40
Earnings	All	0.65	0.15	0.28	51	51
Earnings	White	0.86	0.19	0.28	51	51
Earnings	Hispanic	0.35	0.17	0.10	40	51
Earnings	Black	0.23	0.06	0.28	32	51
Earnings	Asian	0.15	0.45	0.01	19	51
Earnings	AIAN	1.04	0.84	0.34	4	51
Earnings	NHOPI	–	–	–	1	45

Source: IDDA and authors' calculation using IPUMS CPS. Release authorization CBDRB-FY23-0277. The slope are the OLS estimates of unweighted regression across states. The change are computed between 1998 and 2018 for the household-level incomes and between 2005 and 2018 for the individual earnings. The number of observations correspond to states with at least 50 observations in that group in the initial and final years.

remain at the same percentile of the income distribution over time, changes in percentile values do not necessarily align with the income growth experienced by specific people in different parts of the income distribution. The panel dimension of the data underlying IDDA permits the construction of within-person income growth statistics by initial income level, providing more detail on how income growth experiences differ across the income distribution.

Figure 3 shows two approaches to measuring growth in household AGI. Panel (a) plots the values of the 25th, 50th, 75th, and 90th percentiles of the nominal household AGI distribution annually for 1998–2019. Each of the plotted percentiles grows fairly consistently over this period, with higher percentiles growing faster.

Panel (b) shows the median one-year change in nominal household AGI from the prior year, calculated with individuals using the AGI of the household they were part of in each year, by

Table 6: 75/25 percentile ratio changes in CPS vs IDDA across states

Income	Group	Est. slope β	Std. Error	R. Sq.	N_{CPS}	N_{IDDA}
		$\Delta\left(\frac{\text{p75}}{\text{p25}}\right)^{\text{state}}_{\text{CPS}} = \beta\Delta\left(\frac{\text{p75}}{\text{p25}}\right)^{\text{state}}_{\text{IDDA}} + \varepsilon^{\text{state}}$				
HH Gross Income	All	0.69	0.11	0.43	51	51
HH Gross Income	White	0.80	0.13	0.43	51	51
HH Gross Income	Hispanic	0.13	0.65	0.00	33	51
HH Gross Income	Black	0.29	0.73	0.01	30	51
HH Gross Income	Asian				0	51
HH Gross Income	AIAN	5.31	1.00	0.83	7	51
HH Gross Income	NHOPI				0	40
Earnings	All	0.76	0.12	0.46	51	51
Earnings	White	0.95	0.13	0.53	51	51
Earnings	Hispanic	0.03	0.27	0.00	40	51
Earnings	Black	0.36	0.14	0.18	32	51
Earnings	Asian	-0.62	0.42	0.11	19	51
Earnings	AIAN	0.75	0.74	0.25	4	51
Earnings	NHOPI				1	45

Source: IDDA and authors' calculation using IPUMS CPS. Release authorization CBDRB-FY23-0277. The slope are the OLS estimates of unweighted regression across states. The change are computed between 1998 and 2018 for the household-level incomes and between 2005 and 2018 for the individual earnings. The number of observations correspond to states with at least 50 observations in that group in the initial and final years.

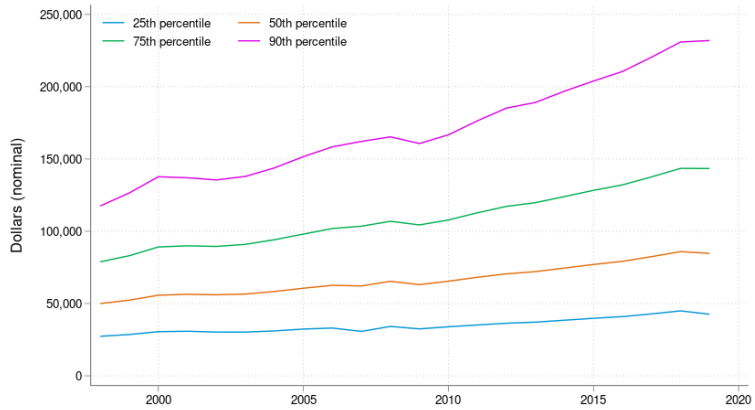
income in the prior year. People who are in the bottom 75 percent of the income distribution in one year do typically experience growth in household AGI between that year and the next, but rather than seeing growth become faster moving up the distribution of initial year income, the bottom three quartiles have similar median one-year growth in nominal dollars (and growth as a share of initial-year income is in fact slower for the higher-income groups). Among people starting in the top decile, median one-year household AGI growth is generally negative (and especially negative during the 2001 and 2007–2009 recessions), likely reflecting mean reversion among people who had temporarily high incomes. For people starting between the 75th and 90th percentiles, growth is typically positive and comparable to that observed for the bottom 75 percentiles, but it turns negative during recessions, mirroring the higher income group.

Together, these approaches suggest that while the incomes of the highest earners are growing

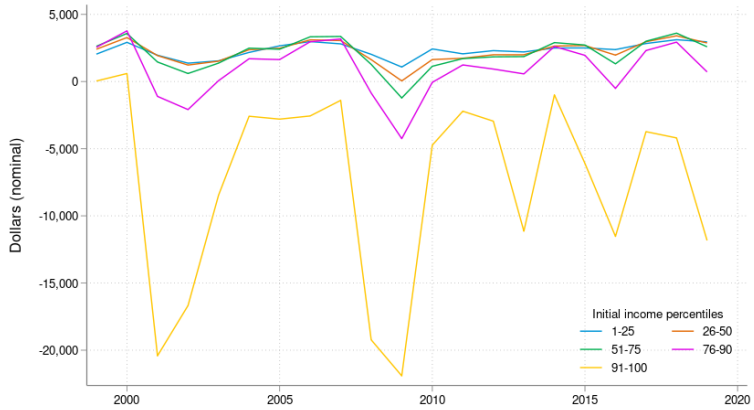
faster than those of the lowest earners, the identities of those high earners change from year to year, and the typical person in a household with income in the top 10 percent in one year will have lower income the following year.

Panels (c) and (d) examine differences in one-year, within-person income growth by race/ethnicity for people with incomes initially in the bottom quartile and top decile of the income distribution, respectively. In panel (c), median growth within all race/ethnicity groups is positive for those starting in the bottom quartile. Non-Hispanic Whites typically see the slowest median growth, perhaps reflecting higher income levels (even among those with relatively low incomes) and less scope to benefit from mean reversion for this group. In contrast, panel (d) shows that non-Hispanic White and non-Hispanic Asian median one-year income growth is least negative (and sometimes positive, especially for non-Hispanic Asians), despite these being the highest income groups. This could suggest that non-Hispanic Whites and non-Hispanic Asians who have high incomes in a given year are less likely to have experienced large one-off increases in income that are unlikely to persist than are members of other race/ethnicity groups.

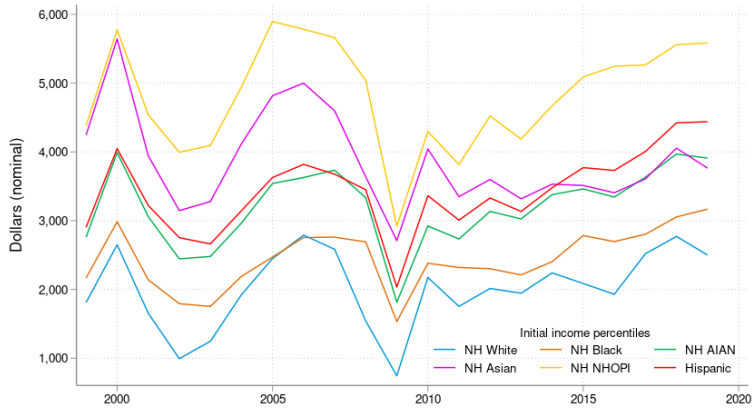
(a) Percentiles of household AGI



(b) Person-weighted median one-year growth in household AGI, by prior year income



(c) One-year growth among people starting in the bottom quartile, by race/ethnicity



(d) One-year growth among people starting in the top decile, by race/ethnicity

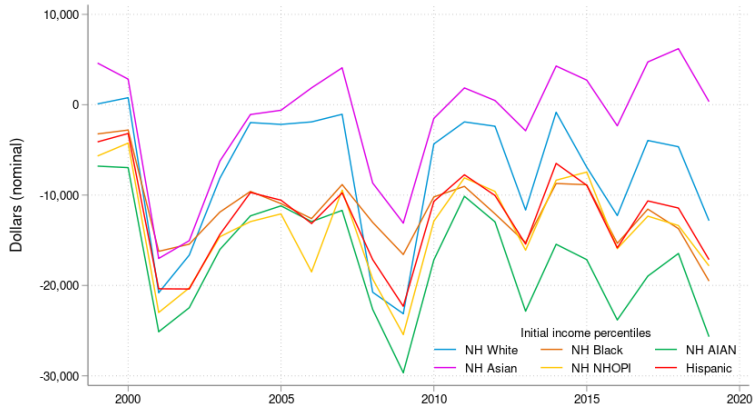


Figure 3: Cross-sectional and panel approaches to measuring income growth

Source: IDDA and authors' calculations.
Note: Values are in nominal dollars. Adjusted gross income (AGI) is aggregated to the household level by summing across tax returns filed from the same address. Growth measures in panel (b) are computed at the individual level based on the income of the household of which each person is a member in each year. Panels (c) and (d) use the within-person income change approach from panel (b) to examine one-year growth by race among people with initial incomes in the bottom quartile and top decile, respectively. Release authorization CBDRB-FY23-0277.

5 Native American and Indigenous Incomes in IDDA

One of the novel dimensions of IDDA is the ability to trace the distribution of Native American incomes not just for the United States as a whole or within any particular state and the District of Columbia, but also within the geographic aggregate of all federally-recognized Native American and Indigenous areas. Here, we show an application that contrasts median earnings of Native American men on and off Native areas.

To answer this question, we combine information from the IDDA module on income percentiles and the IDDA supplement on Native American and Indigenous Areas. The IDDA Native Areas Supplement provides statistics derived from IRS Forms W-2 and 1040 within an aggregate geography that includes all American Indian and Alaska Native/Native Hawaiian areas as delineated in 2017.

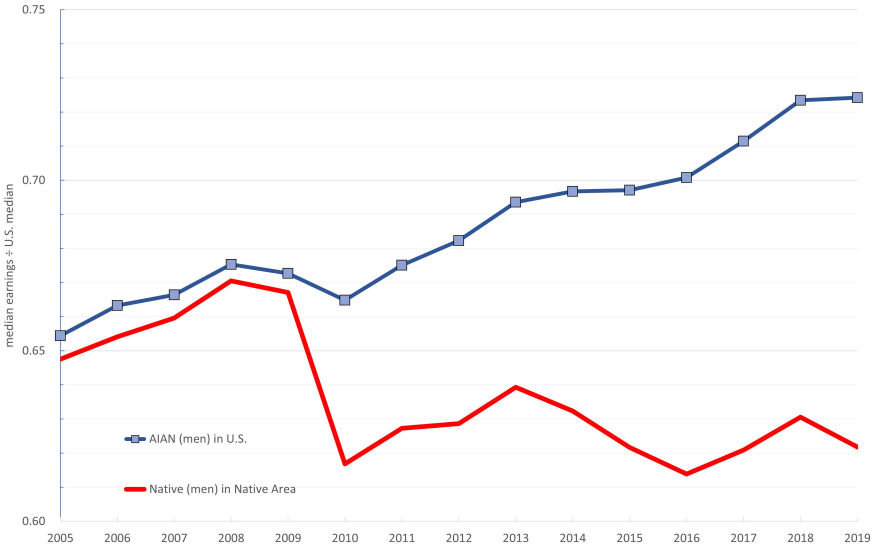


Figure 4: AIAN men median earnings in Native Areas vs. AIAN Men in the US Overall

Source: IDDA and authors' calculation using IPUMS CPS. Release authorization CBDRB-FY23-0277 and CBDRB-FY23-0373.

Figure 4 shows the evolution of median earnings for AIAN men relative to men's median earnings in the U.S. overall. The solid blue line with square markers shows the ratio of median earnings among AIAN men in the U.S. relative median U.S. men earnings. The solid red line shows the ratio of median earnings among Native men in Native Areas relative median U.S. men earnings.

At the beginning of the W-2 sample in 2005, both median AIAN men median earnings in the

U.S. and median AIAN men earnings in Native areas were about two-thirds of median men earnings in the U.S. Leading up the Great Recession, both AIAN men's income were catching up the U.S. median. During the Great Recession, compared to median U.S. men earnings, AIAN men earnings in the U.S. fell slightly while AIAN men earnings in Native areas fell more sharply. Since 2010, AIAN men earnings in the U.S. have continued catching up to median U.S. earnings and reaching almost three-quarters of median U.S. men earnings. In contrast, AIAN men earnings in Native Areas have stopped catching to national incomes and did not make up the ground lost during the Great Recession.

This simple IDDA application highlights nuanced insights that can be drawn from IDDA for informing our understanding of income differences at the intersection of place, income rank, and demographic markers such as race and ethnicity. In the case of AIAN populations in America, this is particularly valuable given the limited availability of granular statistics on Native incomes.

6 Conclusion

Considering income inequality and dynamics within granular demographic and geographic cells can reveal important facts about how inequality evolves in the U.S. This approach also has the potential to enhance our understanding of the functioning of the broader economy, as economists turn to questions beyond how income distributions evolve to questions of what causes the changes we observe. To contribute to this agenda, we developed a set of statistics describing income distributions, income mobility and conditional income growth for communities in the U.S. defined by place (U.S. states) and demographics (race and ethnicity, sex, age, and birthplace). We used the universe of 1040 tax filings from 1998 to 2019 as well as the universe of W-2 filings for 2005 to 2019 to construct measures that reflect earnings distributions in these two populations.

In this paper, we investigate whether this approach yields new patterns of expanding income inequality for groups defined by race and ethnicity, beyond what can be observed in public sources for these groups. We find that major features of U.S.-level earnings distributions, like 75-25 percentile inequality, align well across the measures we construct from tax records and those that could be constructed using publicly available self-reports of earnings from survey data.

We also identify several novel insights. Generally, inequality in the earnings distribution are

imprecisely captured using public data for the highest earnings percentiles. We also find that even standard measures of inequality for race and ethnicity groups at the state level benefit from the large samples available in administrative data. Furthermore, the administrative data allow for construction of a longitudinal record of individual earnings. From these, we can derive measures of earnings dynamics that cannot be constructed elsewhere. We show that earnings dynamics measured with the administrative panel data differ from those inferred from changes in earnings at percentiles of the cross-sections of earnings in survey data.

Finally, to demonstrate the breadth of potential future analysis, we provide an example that illustrates how the evolution of earnings for small race and ethnic groups across places can be studied after constructing granular distributional statistics from the administrative data. Specifically, we find that the earnings trajectories for Native American men residing on and outside of tribal areas diverged sharply after the Great Recessions. Researchers interested in replicating or extending this analysis can access the statistics we calculated through the Income Distributions and Dynamics in America (IDDA) project.

References

Akee, Randall, Maggie R Jones, and Sonya R Porter, "Race matters: Income shares, income inequality, and income mobility for all US races," *Demography*, 2019, *56* (3), 999–1021.

Autor, David, Arindrajit Dube, and Annie McGrew, "The Unexpected Compression: Competition at Work in the Low Wage Labor Market," Technical Report, National Bureau of Economic Research 2023.

Autor, David H, Lawrence F Katz, and Melissa S Kearney, "Trends in US wage inequality: Revising the revisionists," *The Review of economics and statistics*, 2008, *90* (2), 300–323.

Bee, Adam and Joshua Mitchell, "Do Older Americans Have More Income Than We Think?," *Proceedings. Annual Conference on Taxation and Minutes of the Annual Meeting of the National Tax Association*, 2017, *110*, 1–85.

Blanchet, Thomas, Emmanuel Saez, and Gabriel Zucman, "Real-Time Inequality," Technical Report, National Bureau of Economic Research 2022.

Burkhauser, Richard V, Shuaizhang Feng, Stephen P Jenkins, and Jeff Larrimore, "Estimating trends in US income inequality using the Current Population Survey: the importance of controlling for censoring," *The Journal of Economic Inequality*, 2011, *9*, 393–415.

Chetty, Raj, Nathaniel Hendren, John Friedman, Maggie R Jones, and Sonya R Porter, "The Opportunity Atlas: Mapping the Childhood Roots of Social Mobility," Technical Report, National Bureau of Economic Research 2018.

_ , _ , Patrick Kline, and Emmanuel Saez, "Where is the land of opportunity? The geography of intergenerational mobility in the United States," *The Quarterly Journal of Economics*, 2014, *129* (4), 1553–1623.

Flood, Sarah, Miriam King, Renae Rodgers, Steven Ruggles, J. Robert Warren, and Michael Westberry, *Integrated Public Use Microdata Series, Current Population Survey: Version 10.0 [dataset]* Minneapolis, MN: IPUMS 2022. https://doi.org/10.18128/D030.V10.0.

Frank, Mark, "A new state-level panel of annual inequality measures over the period 1916-2005," *Journal of Business Strategies*, 2014, *31* (1), 241–263.

Guvenen, Fatih and Greg Kaplan, "Top income inequality in the 21st century: Some cautionary notes," Technical Report, National Bureau of Economic Research 2017.

_ , _ , Jae Song, and Justin Weidner, "Lifetime earnings in the United States over six decades," *American Economic Journal: Applied Economics*, 2022, *14* (4), 446–79.

Hoffmann, Florian, David S Lee, and Thomas Lemieux, "Growing income inequality in the United States and other advanced economies," *Journal of Economic Perspectives*, 2020, *34* (4), 52–78.

Kopczuk, Wojciech, Emmanuel Saez, and Jae Song, "Earnings inequality and mobility in the United States: evidence from social security data since 1937," *The Quarterly Journal of Economics*, 2010, *125* (1), 91–128.

McKinney, Kevin L., John M. Abowd, and Hubert P. Janicki, "U.S. long-term earnings outcomes by sex, race, ethnicity, and place of birth," *Quantitative Economics*, 2022, *13* (4), 1879–1945.

Morrill, Richard, "Geographic variation in change in income inequality among US states, 1970–1990," *The Annals of Regional Science*, 2000, *34*, 109–130.

Partridge, Mark D, Dan S Rickman, and William Levernier, "Trends in US income inequality: evidence from a panel of states," *The Quarterly Review of Economics and Finance*, 1996, *36* (1), 17–37.

Rinz, Kevin and John Voorheis, "Re-examining Regional Income Convergence: A Distributional Approach," February 2023. CES Working Paper 23-05.

Sommelier, Estelle and Mark Price, "The increasingly unequal states of America," *Washington: Economic Policy Institute*, 2014.

Wagner, D. and M. Layne, "The Person Identification Validation System (PVS): Applying the Center for Administrative Records Research and Applications (CARRA) Record Linkage Software," *Mimeo*, 2014.